## Test 3 – Grade 9

**Name: _____**
**Grade / Class: 9**

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# Section A: English Language & Literature

**1. Reading Comprehension**
**Read the passage and answer the questions:**

***"Leila attended a national science fair to present her project on renewable energy. She built a small solar-powered car and explained the science behind solar cells. The fair allowed her to meet students from different regions, exchange ideas, and gain inspiration for future projects."***

**a) What project did Leila present at the science fair?**
**b) How did the fair help Leila expand her knowledge?**
**c) Mention two benefits of attending such events.**
**d) Write a summary of the passage in four sentences.**

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**2. Grammar & Vocabulary**

**a) Combine using an appropriate conjunction:**
***"The sun had set. The hikers continued their journey."***

**b) Identify adjectives and adverbs in:**
***"The diligent student quickly completed the difficult assignment."***

**c) Correct the error:**
***"Each of the students have submitted their essays."***

**d) Convert to indirect speech:**
***"He said, 'I will attend the science exhibition tomorrow.'"***

**e) Replace underlined word with a synonym:**
***"The mountain scenery was breathtaking."***

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**3. Writing**

**Write a short essay (120–150 words) on:**
**“The Impact of Social Media on Teenagers”**
**Include advantages, disadvantages, and your personal opinion.**

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**4. Literature Analysis**

**a) Identify the theme of a story you studied recently.**
**b) Explain the metaphor in:**
***"Time is a thief that steals our moments."***
**c) Discuss the role of a protagonist and how they influence the plot.**

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# Section B: Mathematics

**1. Algebra**

**a) Solve for x:**
**7x−9=2x+167x - 9 = 2x + 167x−9=2x+16**

**b) Factorize:**
**x2−14x+45x^2 - 14x + 45x2−14x+45**

**c) Simplify:**
**4p2q+7pq2−3p2q4p^2q + 7pq^2 - 3p^2q4p2q+7pq2−3p2q**

**d) Solve simultaneously:**
**2x+3y=132x + 3y = 132x+3y=13, x−y=2x - y = 2x−y=2**

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**2. Geometry & Mensuration**

**a) Find the area of a trapezium with bases 12 cm and 8 cm, height 6 cm.**
**b) A sphere has radius 7 cm. Find its volume.**
**c) Find distance between points (3, -4) and (7, 2).**

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**3. Trigonometry**

**a) Find sin θ if cos θ = 0.6 and θ is acute.**
**b) Verify identity: 1+cot2x=csc2x1 + cot^2x = csc^2x1+cot2x=csc2x**
**c) A tree casts a shadow 9 m long. Angle of elevation is 30°. Find the height of the tree.**

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**4. Ratio, Proportion & Percentage**

**a) A jacket costs $180. It is sold at 20% discount. Find selling price.**
**b) If 5 notebooks cost $15, find the cost of 18 notebooks.**
**c) The population of a town decreased from 120,000 to 114,000. Calculate the percentage decrease.**

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**5. Statistics & Probability**

**a) Marks of 12 students: 8, 12, 10, 9, 11, 12, 10, 13, 9, 12, 11, 10. Find mean, median, mode.**
**b) Construct a frequency table for the above data.**
**c) A bag contains 2 green, 3 red, and 5 yellow balls. Probability of picking a red or yellow ball?**

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# Section C: Science

**1. Biology**

**a) Describe the circulatory system in humans.**
**b) Explain mitosis and its stages.**
**c) Difference between exocrine and endocrine glands.**
**d) Two adaptations of desert animals.**
**e) Define food web with an example.**

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**2. Physics**

**a) A train moves 360 m in 18 s. Calculate average speed.**
**b) Define kinetic energy and give one example.**
**c) Difference between conduction, convection, and radiation.**
**d) Angle of incidence = 30°. Find angle of reflection.**
**e) State and explain Newton's third law of motion.**

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**3. Chemistry**

**a) Identify as element, compound, or mixture: Helium, Salt, Sugar water.**
**b) Balanced equation: Magnesium reacts with hydrochloric acid.**
**c) Explain ionic and covalent bonds.**
**d) Define acid with one example.**
**e) Law of conservation of mass – explain with one example.**

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**4. Environmental Science**

**a) List two major water pollutants and their effects.**
**b) Suggest three methods to save electricity at home.**
**c) Explain global warming and its consequences.**
**d) How does afforestation help biodiversity?**

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## Section D: Social Studies

**1. History**

**a) Causes and effects of the Russian Revolution.**
**b) Importance of the Renaissance in European history.**
**c) Major allied countries during WWI.**

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**2. Geography**

**a) Define climate zones and give one example.**
**b) Name the largest lake in Africa.**
**c) Effects of soil erosion on agriculture – mention two.**

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**3. Civics / Political Science**

**a) Difference between dictatorship and democracy.**
**b) Two responsibilities of citizens in a federal government.**

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**4. Economics**

**a) Explain factors affecting demand.**
**b) Difference between fixed and variable costs.**
**c) Importance of budgeting for households.**

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## Section E: ICT / Computer Studies

**1. Computer Basics**

**a) Explain difference between software and hardware with examples.**
**b) Name one input device and one output device.**

**2. Applications**

**a) Which spreadsheet function finds the average?**
**b) Two advantages of using databases in schools.**

**3. Programming Basics**

**a) Define a function in Python and give an example.**
**b) Write a Python code to print the first 10 odd numbers.**

**4. Cyber Awareness**

**a) Explain ransomware and one method to prevent it.**
**b) List two ways to secure Wi-Fi networks.**

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## Section F: Islamiyat

**1. Quran & Hadith**

**a) Name two values emphasized in the Quran.**
**b) Narrate a Hadith promoting respect for parents.**

**2. Aqidah / Belief**

**a) List the five pillars of Islam.**
**b) Explain the concept of Qadr (Divine Decree).**

**3. Fiqh / Worship**

**a) Significance of Salah in a Muslim's daily life.**
**b) Explain rules of fasting during Ramadan.**

**4. Islamic Ethics & Morality**

**a) How does Islam encourage fairness in trade?**
**b) Suggest two ways Muslims are encouraged to protect the environment.**

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## Section G: Arts & Music

**1. Visual Arts**

**a) Explain cross-hatching in drawing.**
**b) Name three primary colors and their secondary color combination.**

**2. Music**

**a) Define rhythm and tempo in music.**
**b) Give an example of a percussion and a string instrument.**

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## Section H: Health & Physical Education

**1. Physical Fitness**

**a) Name two exercises to strengthen core muscles.**
**b) Differentiate between aerobic and anaerobic exercises.**

**2. Health & Hygiene**

**a) Why is water sanitation important?**
**b) Two ways to prevent vector-borne diseases.**

**3. Safety & Environment**

**a) State one rule for chemical storage in labs.**
**b) Mention one method to prevent fire accidents at home.**